

Theorem (Young's inequality). *Let $a, b \geq 0$ and let $p, q > 1$ satisfy $\frac{1}{p} + \frac{1}{q} = 1$. Then*

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q} \quad (1)$$

and equality holds iff either $a = b = 0$ or $a, b > 0$ and $a^p = b^q$ (equivalently $a^{p-1} = b^{q-1}$).

Proof. 1. Degenerate cases. If $a = 0$, then $ab = 0$ and the right-hand side of (??) equals $\frac{0}{p} + \frac{b^q}{q} \geq 0$, so (??) holds. The same argument works when $b = 0$. When $a = b = 0$ both sides are 0, hence equality occurs.

2. Conjugate exponents. Set

$$\lambda := \frac{1}{p}, \quad \mu := \frac{1}{q}.$$

Because $p, q > 1$ and $\frac{1}{p} + \frac{1}{q} = 1$, we have

$$0 < \lambda, \mu < 1, \quad \lambda + \mu = 1. \quad (2)$$

3. Convexity of the exponential. The function $f(t) = e^t$ is convex on $\mathbb{R}$. For any $X, Y \in \mathbb{R}$ and $\lambda \in [0, 1]$ Jensen's inequality (two-point case) gives

$$e^{\lambda X + \mu Y} \leq \lambda e^X + \mu e^Y, \quad (3)$$

with equality iff $X = Y$ (when $\lambda, \mu > 0$).

4. Choice of X and Y . Assume $a > 0$ and $b > 0$ (the degenerate cases have been settled). Define

$$X := p \ln a, \quad Y := q \ln b.$$

5. Left-hand side of (??). Using (??),

$$\lambda X + \mu Y = \frac{1}{p}(p \ln a) + \frac{1}{q}(q \ln b) = \ln a + \ln b = \ln(ab),$$

hence

$$e^{\lambda X + \mu Y} = e^{\ln(ab)} = ab. \quad (4)$$

6. Right-hand side of (??). Again by (??),

$$\lambda e^X + \mu e^Y = \frac{1}{p}e^{p \ln a} + \frac{1}{q}e^{q \ln b} = \frac{a^p}{p} + \frac{b^q}{q}. \quad (5)$$

7. Conclusion of the inequality. Substituting (??) and (??) into (??) yields

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q},$$

which is exactly (??).

8. Equality condition. In (??) equality holds precisely when $X = Y$, i.e.

$$p \ln a = q \ln b \iff a^p = b^q.$$

Together with the degenerate case $a = b = 0$, the equality cases are those stated in the theorem.

Thus Young's inequality is proved, together with its equality conditions. $\square$
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